INDUCTANCE OF CIRCULAR LOOP file: circular.mcd

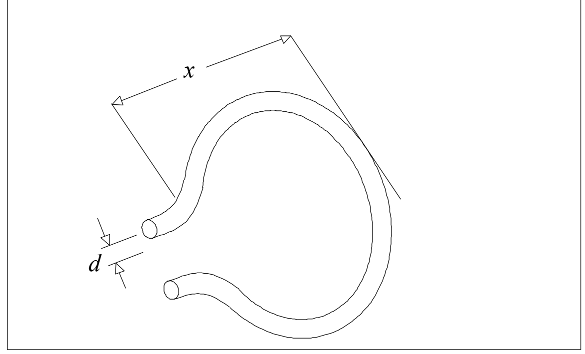
Formulas included in this spreadsheet:

Inductance of circular wire loop	LCIRC()
Impedance magnitude of inductor at one frequency	XLF()
Impedance magnitude of inductor as seen by rising edge	XLR()

Variables used:

d Diameter of wire (in.)

x Diameter of wire loop (in.)



Inductance of a circular wire loop

circular

Inductance of wire loop (H):

$$LCIRC(d,x) := 1.56 \cdot 10^{-8} \cdot x \cdot \left(\ln\left(\frac{8 \cdot x}{d} \right) - 2 \right)$$

A loop of 24-gauge wire the size of the loop between your thumb and forefinger has about 100 nH of inductance.

$$LCIRC(.01, 1.3) = 1.003 \times 10^{-7}$$

Changing the wire diameter from AWG 24 to AWG 14 makes little difference. The log function is rather insensitive to wire size.

$$LCIRC(.1, 1.3) = 5.363 \times 10^{-8}$$

Impedance magnitude of inductor at frequency f (Ω):

l Inductance (H)

f Frequency (Hz)

$$XLF(l,f) := 2 \cdot \pi \cdot f \cdot l$$

The impedance, at 100 MHz, of a
100-nH inductor is 62 Ω.

$$XLF\left(100 \cdot 10^{-9}, 10^{8}\right) = 62.832$$

Impedance magnitude of inductor as seen by rising edge (Ω):

l Inductance (H)

tr 10-90% rise time (s)

$$XLT(l,tr) := \frac{\pi \cdot l}{tr}$$

The impedance, as seen by a 5-ns rising edge,
of a 100-nH inductor is 62 Ω.

$$XLT\left(100 \cdot 10^{-9}, 5 \cdot 10^{-9}\right) = 62.832$$